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Effects of underwater breath-holding training on athletes' lung capacity, heart rate, blood pressure, and lung CT imaging

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Abstract

Background: The aim of this study was to explore the effects of underwater breathing-holding training on the cardiovascular function and swimming performance of athletes.

Methods: Sixty swimming athletes from sports colleges are separated into an experimental group and a control group, with 30 athletes in each one. Two groups undergo 8 weeks of swimming training with a total of 32 lessons. The control group athletes receive routine swimming training. The experimental group adds breath-holding training during swimming training. Before and after training, cardiopulmonary function tests are conducted on the two groups, including heart rate, lung capacity, blood pressure, and breath-holding time. Meanwhile, the maximum suction pressure, suction flow rate, and maximum ventilation volume of the athletes are tested before and after the experiment. In addition, the study also adds a 50m freestyle swimming test to explore the swimming performance of two groups before and after training.

Results: After 8 weeks, the overall improvement effect of the experimental group was more significant. In the cardiopulmonary function test, the lung capacity of the experimental group athletes increased from 3.24 ± 1.69 L to 4.81 ± 1.93 L, with statistical significance ($P < 0.05$). For the control group, the lung capacity before and after the experiment did not exhibit statistically significant difference ($P > 0.05$). Meanwhile, the heart rate of the experimental group athletes decreased from 58.61 ± 9.32 beats/minute to 56.39 ± 8.28 beats/minute, with a P value of only 0.1668, lower than that of the control group's 0.4412. In addition, the breath-holding time of the experimental group increased from 54.36 ± 2.05 s before the experiment to 60.15 ± 3.28 s ($P < 0.05$). In the index tests of maximum suction pressure, suction flow rate, and ventilation volume, the experimental group had statistically significant differences before and after the experiment ($P < 0.05$). The control group only had statistical significance in maximum suction pressure and suction flow rate. In addition, the 50m freestyle swimming time of the experimental group athletes was reduced from 25.52 ± 0.59 s to 25.20 ± 0.66 s ($P < 0.05$). The free swimming performance of the control group before and after training was not statistically significant ($P > 0.05$). The results of lung CT examination showed that the lung function of the experimental group athletes significantly improved. The lung imaging images showed larger lung volume, unobstructed airway, clear alveolar structure, and no obvious lung lesions or atrophy.

Conclusions: The combination of breath-holding training and routine training in swimming training can improve the cardiovascular function of athletes, enhance their respiratory function, and ultimately enhance their swimming performance. This training method provides a scientifically effective training strategy for swimmers.

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